DECODELABS • INDUSTRIAL TRAINING KIT • BATCH 2026

DATA ANALYTICS PROJECT 4

Data Visualization & Business Storytelling

Turning 1,200 order records into clear, decision-ready insights.

1,200	1.26M	3,535	1,189
Orders analysed	Recorded order value	Units ordered	Unique customers

Prepared from: Dataset for Data Analytics-4.xlsx
Project brief: Data Analytics Project 4 – Data Visualization
Analysis period: 1 January 2023 to 30 June 2025

A practical visualization report applying the Architect, Editor and Storyteller disciplines from the DecodeLabs project brief.

1. Executive summary

This analysis covers **1,200 orders** recorded between **01 January 2023** and **30 June 2025**. The dataset records **1,264,761.96** in total order value across **3,535 units**. Average order value is **1,053.97**, while the median is **823.62**. The currency is not specified in the source file, so values are reported in dataset currency.

1,053.97	823.62	11	38.8%
Average order value	Median order value	Customers with 2+ orders	Shipped or delivered

What stands out: Chair and Printer have the highest recorded order values (about 195,620.11 each). The most frequently recorded status is Cancelled (250 orders), while Shipped and Delivered together account for 466 orders (38.8%). Instagram is the largest referral source by order count. These are descriptive findings, not proof of causation.

Important interpretation: TotalPrice is calculated as Quantity × UnitPrice and is present for every order regardless of status. Therefore, the sum is best described as *recorded order value*, not realized revenue or cash collected. Cancelled and returned orders must be excluded or adjusted before reporting net sales.

Recommended management focus: investigate cancellation and return drivers, validate fulfilment outcomes, and review product-level performance before making inventory or marketing allocation decisions.

2. Dataset overview and methodology

The workbook contains one worksheet (Sheet1), with 1,200 rows and 14 columns. Each row represents an order record. Fields include order/customer identifiers, date, product, quantity, unit price, shipping address, payment method, order status, tracking number, items in cart, coupon code, referral source and total price.

Field group	Columns	Analytical use
Identifiers	OrderID, CustomerID, TrackingNumber	Record uniqueness and customer-level counts
Time & product	Date, Product	Trend and product comparisons
Order value	Quantity, UnitPrice, TotalPrice	Order value and unit-volume analysis
Operations	PaymentMethod, OrderStatus	Payment mix and fulfilment outcomes
Marketing	CouponCode, ReferralSource	Coupon recording and acquisition-source mix
Other	ShippingAddress, ItemsInCart	Context; limited use in this report

Method: imported the workbook, checked missing values and duplicates, validated the TotalPrice arithmetic against Quantity × UnitPrice, and grouped records by month, product, order status, payment method and referral source. Monetary calculations retain the source precision. No imputation or fabricated records were introduced.

Scope and limitations

The file does not specify currency, profit/cost, customer demographics, geographic region, marketing spend, coupon redemption rules, or timestamps for shipment and delivery. It also does not explain whether each row is a unique transaction or whether status categories represent the latest status. Consequently, this report does not estimate profit, customer lifetime value, campaign ROI, delivery speed or geographic performance. 2025 contains only six months and is not directly comparable with full-year 2023 or 2024.

3. Data quality and preparation

Check	Result	Interpretation
Rows / columns	1,200 / 14	One worksheet analysed
Missing values	309 in CouponCode; 0 elsewhere	25.8% of coupon entries are blank
Exact duplicate rows	0	No exact duplicate records found
Duplicate OrderID	0	OrderID is unique in this file
TotalPrice validation	0 mismatches above 0.01	All records reconcile to Quantity × UnitPrice within 0.01
Date range	01 Jan 2023 – 30 Jun 2025	30 months represented

Data-quality decisions

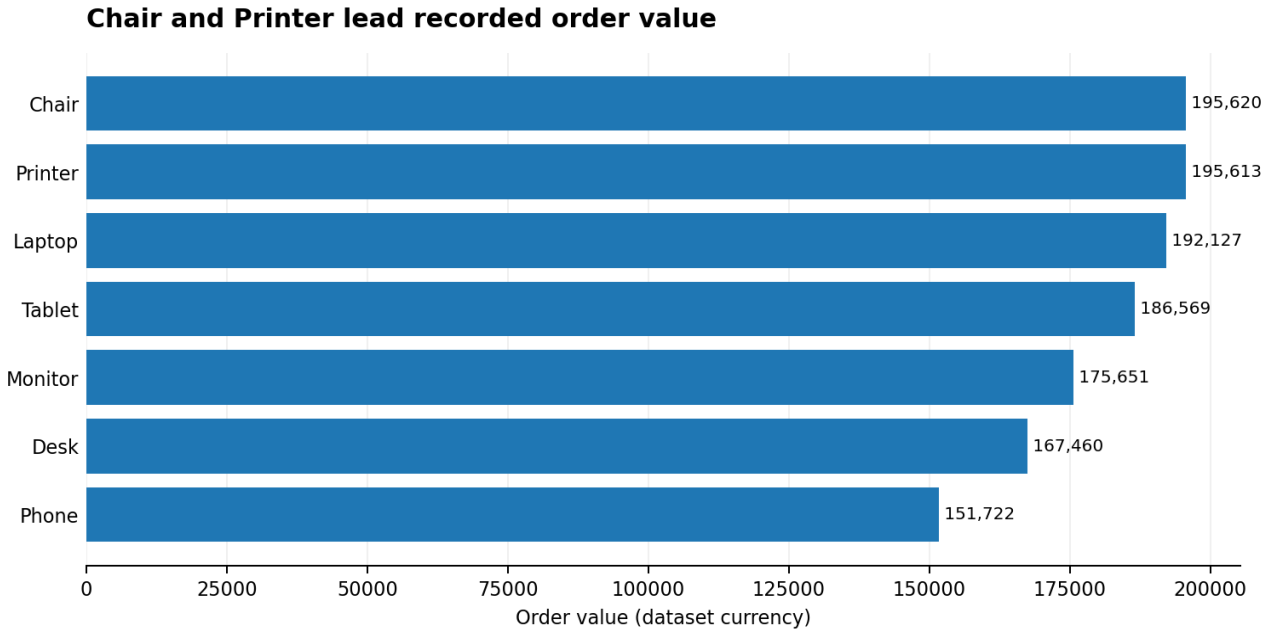
• CouponCode blanks were retained as “No coupon recorded” for the coupon mix. A blank does not prove that no coupon was used; it may also indicate missing capture.
• No rows were removed because no exact duplicates or arithmetic inconsistencies were detected.
• CustomerID repeats were retained: 1,189 unique customer IDs appear across 1,200 orders, and 11 customers have more than one order. This suggests a small repeat-order group in the supplied file, but the sample period and synthetic-looking IDs limit broader conclusions.
• Shipping addresses and tracking numbers were not used in charts because they are operational identifiers and do not support a defensible geographic or delivery-time analysis here.

A key analytical caution

The five order statuses are nearly balanced. Since Cancelled and Returned orders are included in TotalPrice, the dataset’s total value should not be described as sales revenue without a business rule for recognizing, reversing or excluding these statuses.

4. Product performance

Chair has the highest recorded order value at 195,620.11, narrowly ahead of Printer at 195,612.61. Phone has the lowest total at 151,722.39. The gap between the top and bottom products is 22.4% of the top product's value.



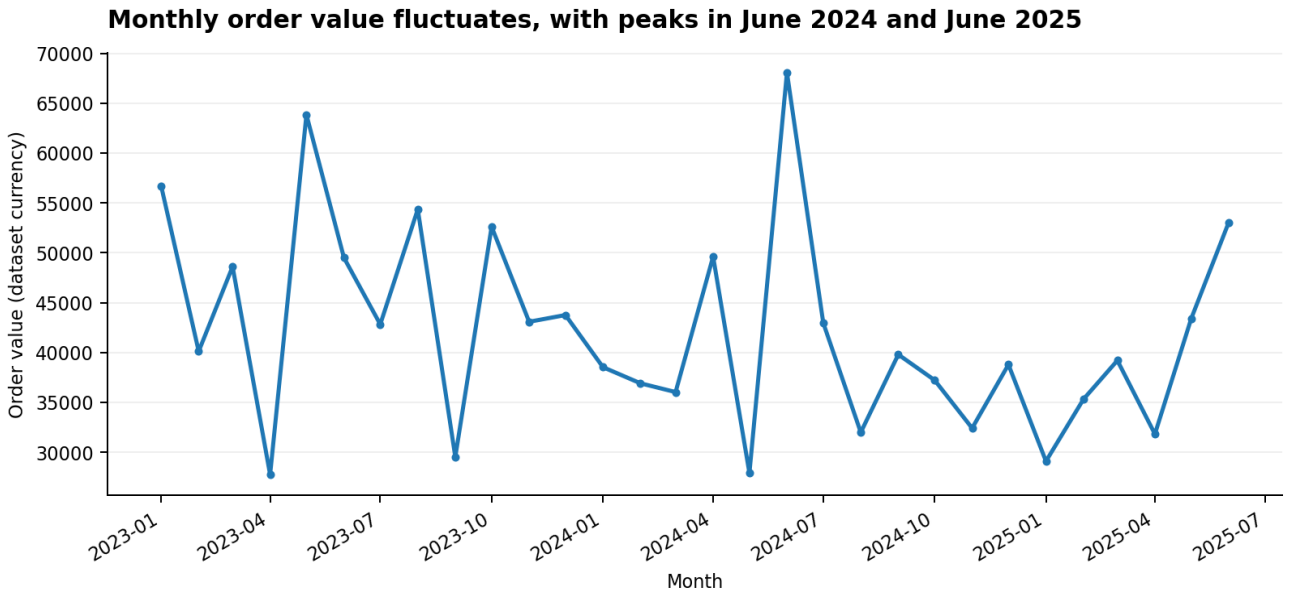
Product	Orders	Units	Recorded order value
Chair	178	562	195,620.11
Printer	181	542	195,612.61
Laptop	173	535	192,126.56
Tablet	179	497	186,568.95
Monitor	163	480	175,651.41
Desk	170	508	167,459.93
Phone	156	411	151,722.39

Interpretation: product totals are relatively close, so no single product dominates the recorded order value. Chair and Printer are the leading categories by value; however, the dataset contains no product costs or margin data. High order value alone does not establish that a product is more profitable.

Action: compare product-level gross margin, stock availability, return rates and customer demand before increasing inventory for the top-value categories.

5. Time trend analysis

Monthly recorded order value varies over the 30-month period. The highest month is June 2024 (68,068.54 across 53 orders); the lowest is April 2023 (27,751.71 across 31 orders). The chart shows repeated peaks rather than a smooth, sustained upward trajectory.



Full-year comparison

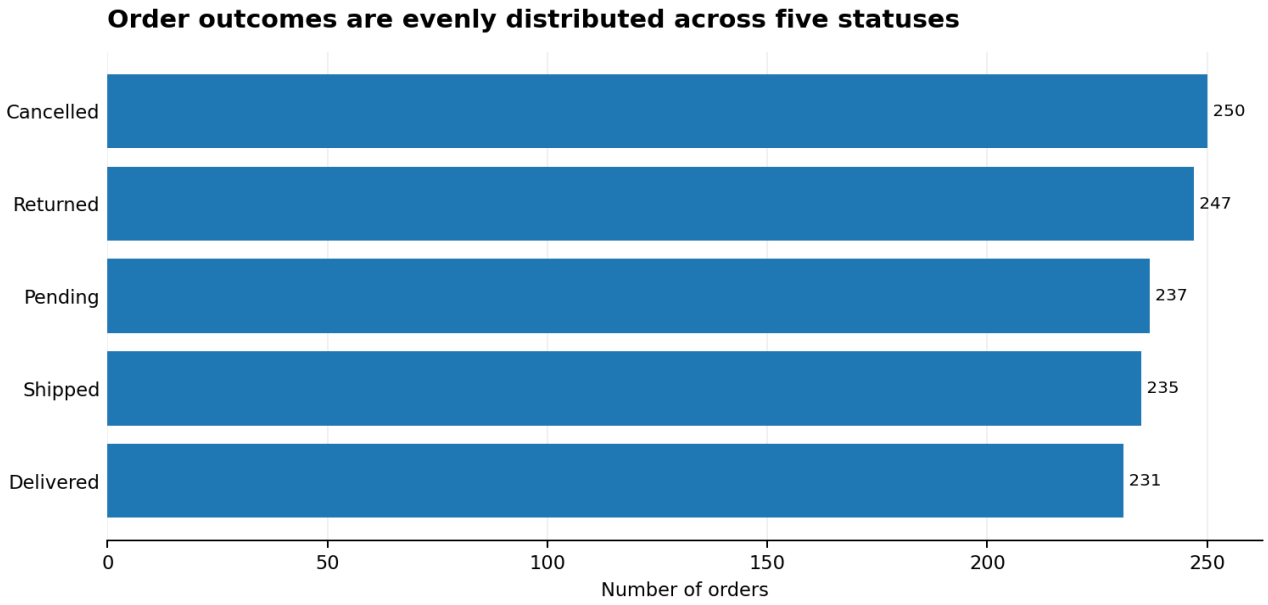
Year	Months included	Orders	Recorded order value
2023	12	510	552,643.24
2024	12	459	480,235.87
2025	6	231	231,882.85

2024 recorded order value was -13.1% higher than 2023. This is a valid full-year comparison because both years contain 12 months. 2025 is partial (January–June) and should not be compared with either full year without normalizing for the shorter period and seasonality.

Action: review what operational or promotional factors coincided with the stronger months (especially June 2024 and June 2025). The dataset alone cannot establish why those peaks occurred.

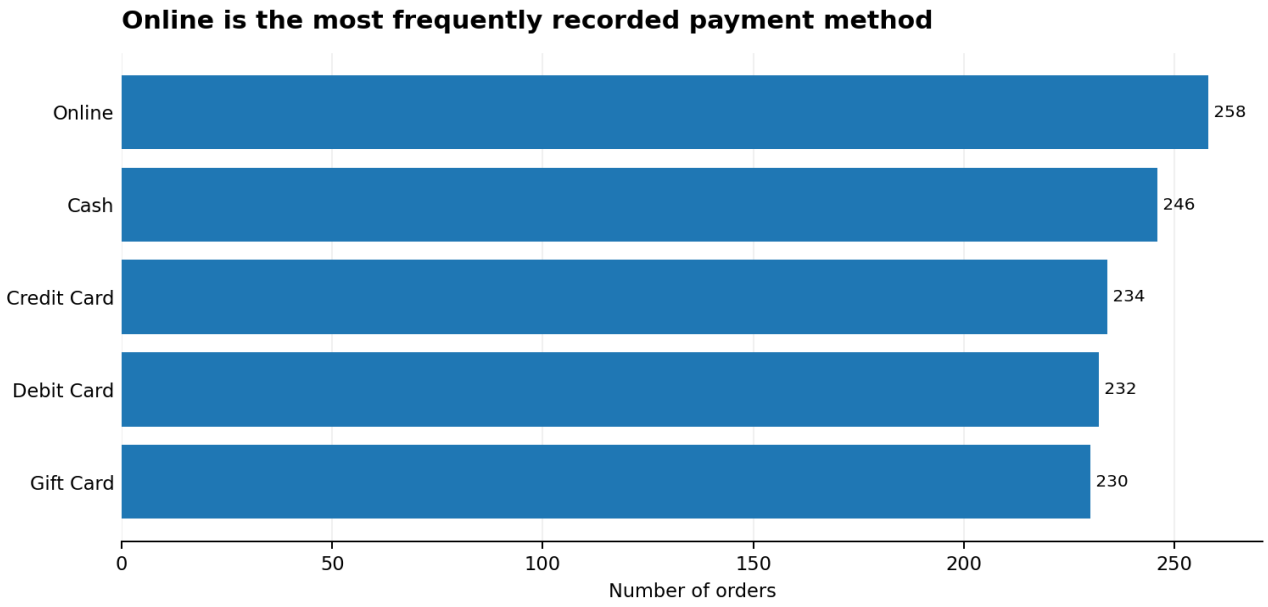
6. Order outcomes and payment methods

The order-status mix is unusually even: no single status represents more than one-fifth of the records. Cancelled is the largest status group, while Delivered is the smallest. Shipped and Delivered together account for less than half of orders.



Status	Orders	Share
Cancelled	250	20.8%
Returned	247	20.6%
Pending	237	19.8%
Shipped	235	19.6%
Delivered	231	19.2%

Payment method mix

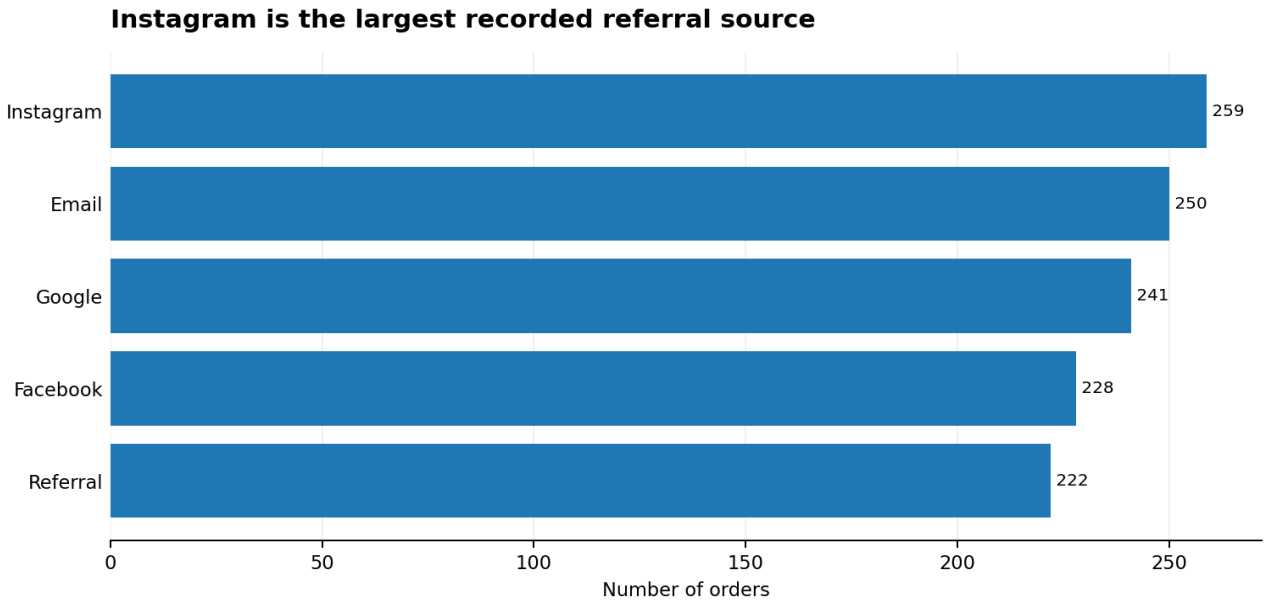


Online is the most frequently recorded payment method (258.0 orders, 21.5%). Gift Card is least frequent (230.0 orders). The differences are modest; payment method is not evidence of customer preference beyond the observed records.

Action: investigate cancellation and return reasons by product, payment method and referral source. Confirm whether “Shipped” is a completed or in-transit state and define how cancelled/returned order value should be handled in financial reporting.

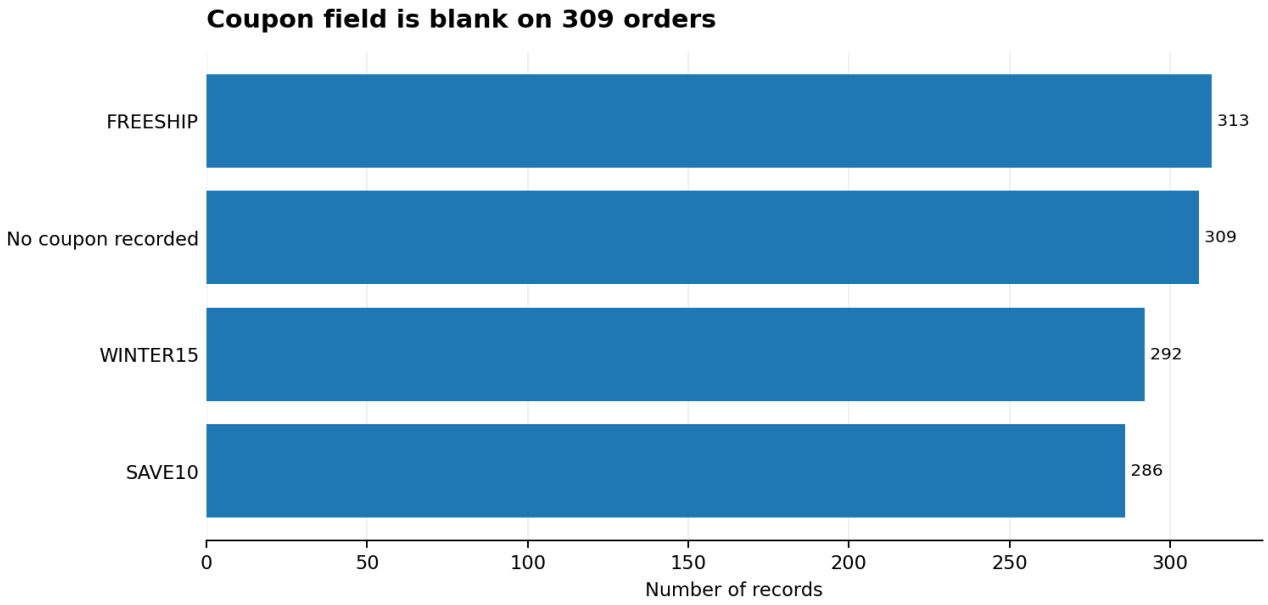
7. Referral and coupon patterns

Instagram is the largest recorded referral source by order count (259), followed by Email (250) and Google (241). Referral is the smallest source (222). Counts are relatively close, and no marketing spend or conversion denominator is supplied, so the data cannot identify the most cost-effective channel.



Referral source	Orders	Share
Instagram	259	21.6%
Email	250	20.8%
Google	241	20.1%
Facebook	228	19.0%
Referral	222	18.5%

Coupon field coverage



The CouponCode field is blank for 309 orders (25.8%). Among populated values, FREESHIP appears most often (313), followed by WINTER15 (292) and SAVE10 (286). Because blank entries may mean either no coupon or missing data, coupon performance should not be inferred from counts alone.

Action: improve coupon-code capture and add campaign spend, impressions, clicks and conversions to measure channel and coupon effectiveness.

8. Business story using the SCR framework

Situation

The business recorded 1,200 orders and 1,264,761.96 in total order value over January 2023–June 2025. Product values are spread across seven categories, with Chair and Printer slightly ahead. Referral-source and payment-method volumes are also relatively balanced.

Complication

497 orders (41.4%) are marked Cancelled or Returned, while only 466 (38.8%) are marked Shipped or Delivered. This raises a fulfilment and revenue-recognition question. In addition, coupon information is blank for 309 records, limiting campaign analysis.

Resolution

Prioritize an order-outcome investigation: segment cancellations and returns by product, payment method, referral source and month; validate status definitions; and calculate net realized sales using a documented accounting rule. Improve coupon data capture and add cost and marketing-spend fields before making profitability or ROI decisions.

Prioritized recommendations

Priority	Recommended action	Why it matters
1 – High	Audit Cancelled and Returned orders; identify root causes and establish a net sales definition.	Prevents overstating revenue and surfaces operational leakage.
2 – High	Build a status dashboard by month, product, payment method and referral source.	Shows where unsuccessful orders are concentrated.
3 – Medium	Improve coupon-code capture and distinguish “no coupon” from “data missing”.	Makes promotion analysis more reliable.
4 – Medium	Add product cost, shipping cost, discounts/refunds and marketing spend.	Enables margin, net revenue and channel ROI analysis.
5 – Medium	Review stock and margin for Chair and Printer; do not allocate inventory or ad budget on sales volume alone.	Balances demand with profitability and return risk.

9. Conclusion and next steps

The dataset provides a useful foundation for descriptive data visualization. It shows 1,200 order records, relatively balanced product and channel mixes, month-to-month order-value fluctuations, and a material share of orders recorded as cancelled or returned. The strongest decision signal is operational: the business should understand unsuccessful order outcomes before treating recorded order value as realized sales.

The analysis follows the three disciplines in the project brief:

The Architect – Choose with Purpose: bar charts compare categories; a line chart shows monthly movement; and axes are kept honest and readable.
The Editor – Edit with Precision: charts use restrained styling, limited visual clutter and direct value labels.
The Storyteller – Tell a Story: each chart has a conclusion-led title, and the findings are organized using Situation, Complication and Resolution.

Next analytical steps

1. Obtain a data dictionary and confirm currency, order-status definitions and whether TotalPrice includes discounts, tax or shipping.
2. Reconcile cancelled and returned orders to refunds and payment records.
3. Add product cost and marketing expenditure to measure margin and acquisition efficiency.
4. Create a repeat-customer and cohort analysis using a longer, validated customer history.
5. Refresh the dashboard monthly and track the same KPIs consistently.

Final takeaway

The immediate priority is not simply to sell more of the highest-value products. It is to understand why so many recorded orders do not reach a Delivered status, and to ensure that management decisions are based on net, validated business outcomes.

Appendix: key metric definitions

Metric	Definition used in this report
Recorded order value	Sum of TotalPrice across all rows, including all order statuses. Not net realized revenue.
Average order value	Mean TotalPrice per order record.
Order share	Category order count divided by 1,200 total records.
Shipped or delivered	Count of records whose OrderStatus is Shipped or Delivered.
Cancelled or returned	Count of records whose OrderStatus is Cancelled or Returned.
Repeat customer	CustomerID appearing on more than one order record.

Source: Dataset for Data Analytics-4.xlsx and DecodeLabs Data Analytics Project 4 instruction document. Prepared for educational project submission.